Data Map — kaun sa data kahan rakha hai

Build 240

DATA	KAHAN RAKHA HAI	KAUN PADHTA / LIKHTA	NOTE
Students, admins, institutes, batches	**Firebase Firestore**	Student/Admin app (SDK), Worker (REST)	institutes/{id}/batches, items, roster ... ; students, admins
Tests aur questions (tests, qchunks)	**Firebase Firestore**	Admin likhta, Student padhta	tests/{id}/qchunks = bade tests ke tukde
Student ke results / records	**Firebase Firestore**	Student likhta, Admin padhta	studentRecords (sabse zyada reads yahin se)
Question Bank (master)	**Firebase Firestore**	Admin app	questionBank ~7,550 docs; poora padhna mehnga
Question Bank (tez copy)	**Cloudflare KV + phone IndexedDB**	Worker /bank, Admin app	Pehle yahin se, fail ho to peela banner (v235)
Version marker (read-diet)	**Firebase Firestore**	Admin likhta, apps padhte	institutes/{id}/ver/main: cache valid hai ya nahi
Login: Admin / Owner	**Firebase Authentication**	Email + password	Owner = OWNER_EMAILS (Worker secret)
Login: Student	**Firebase Auth (anonymous) + Firestore**	Mobile + PIN (Firestore se check)	Server hashing (Phase 4) code me hai par OFF
Uploaded files (PDF, photo, video)	**Google Drive (institute ka)**	Admin app, Google OAuth	Upload ab Drive se; Storage SDK / R2 band
Push notification tokens + bhejna	**Firebase Cloud Messaging**	Worker /broadcast, /alert	App me @capacitor-firebase/messaging
Owner alerts ka history	**Cloudflare KV (key: alerts)**	Worker, Owner App 'Alerts'	Aakhri 60 + Telegram copy
Reads Monitor ke logs	**Cloudflare D1 (READS_DB)**	Tracker -> /readlog -> Owner 'Reads'	14 din baad auto-saaf. Console ke reads nahi ginta
Exam registration responses	**Google Forms (Apps Script)**	Admin app pull karta hai	exam-registration.js
App ka code / APK / versions	**GitHub (repo, Pages, Releases)**	Aap zip upload, Actions banate	snaptestpro.github.io, app-version.json
Offline cache (phone par)	**localStorage + IndexedDB + sw.js**	Har app khud	Records/students list, bank copy, tests meta

Rang: peela = Firebase, narangi = Cloudflare, hara = Google (Drive/Forms), grey = GitHub, neela = phone ke andar.

Dhyan: Firebase Console me collection kholne se bhi reads lagte hain (Usage me gine jaate hain, hamare Reads tab me nahi).